

Executive Summary: Statistical Testing Results

TikTok Claims Classification Project

The TikTok data team aims to create a machine learning model to classify user submission claims effectively. As part of this initiative, the team performed a hypothesis test to examine the relationship between `verified_status` and `video_view_count`.

Key Insights

The analysis reveals a difference in the number of views between TikTok videos posted by verified accounts and those posted by unverified accounts.

These findings suggest potential underlying behavioral differences between the two groups. Investigating the root cause of these differences could yield valuable insights. For example:

- Do unverified accounts tend to post more engaging content? If so, is that content based on claims or opinions?
- Alternatively, are unverified accounts associated with spam bots that artificially boost view counts?

Exploring these questions could help uncover the driving factors behind the observed disparities.

Details

The TikTok data team explored the relationship between `verified_status` and `video_view_count` using two approaches. First, the team analyzed the mean values of `video_view_count` for each `verified_status` group in the sample data. The analysis revealed that unverified accounts had an average of 265,663 views, compared to 91,439 views for verified accounts.

```
verified_status
not verified    265663.785339
verified       91439.164167
Name: video_view_count, dtype: float64
```

Next, a two-sample hypothesis test was conducted. Consistent with the initial findings from the mean values, this statistical analysis confirmed that the observed difference in the sample data reflects a true difference in the population means.

Next Steps

The team recommends proceeding with the development of a regression model focused on `verified_status`. This model will provide valuable insights into user behavior within the group of verified accounts. The findings from this analysis can then serve as context for interpreting the results of a claim classification model to be developed later.